An Enhanced Li-ion Battery Model with Optimized OCV Estimation

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***Abstract*—Accurate open-circuit voltage (OCV) estimation is critical for battery management systems (BMS) in ensuring both performance and safety of lithium-ion batteries. This study compares two approaches: a dynamic Kalman Filter (KF) that integrates state-of-charge (SoC) into its state vector for adaptive, real-time estimation, and an offline Polynomial Regression (PR) model. Data were collected from controlled charge–discharge cycles of 3500 mAh cells at 25°C, including experiments on an EVE-INR18650/35V platform. While both models achieve high accuracy as measured by metrics such as mean square error (MSE), the KF consistently yields near-zero errors through adaptive noise tuning, demonstrating superior robustness against measurement noise. In contrast, the simpler PR model, despite good generalization across SoC levels, exhibits relatively higher errors. These results highlight the necessity for adaptive, real-time methods in modern BMS and pave the way for more resilient battery management strategies.**

***Index Terms*—Battery Management Systems, State of Health, State of Charge, Process Noise Covariance, Measurement Noise Covariance**

I. Introduction

Lithium-ion batteries are widely used in portable electronics, electric vehicles (EVs), and large-scale energy storage systems. Accurate open circuit voltage (OCV) estimation is essential to ensure battery reliability, safety, and performance [1], and understanding the relationship between OCV and state of charge (SoC), battery management systems (BMS) can monitor SoC and state of health (SoH), enabling informed decisions for optimal operation [2].

This study aims to develop and evaluate an adaptive, real-time OCV estimation framework by comparing a dynamic Kalman Filter (KF) that integrates state-of-charge (SoC) for adaptive tuning with a conventional Polynomial Regression (PR) model, specifically comparing each method's accuracy, consistency, and robustness using metrics such as mean square error (MSE) and root mean square error (RMSE) from per-cell train-test splits across multiple charge–discharge cycles, and assessing the KF's capability to handle real-time dynamic conditions through adaptive tuning of its noise covariances (Q and R) versus the static curve-fitting approach of PR.

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Few studies directly compare Kalman Filters to polynomial regression under realistic noise conditions or address cell-to-cell variations and dynamic tuning strategies [3]–[5].

This study makes several key contributions through enhanced adaptive filtering where we extend KF by incorporating SoC into the state vector and propose dynamic tuning strategies for noise covariances based on real-world sensor variances [1], [3], and comparative analysis where per-cell train-test splits provide detailed KF vs. PR comparison with overfitting analysis.

II. Methodology

A. Data Collection

Battery datasets were collected from multiple charge-discharge cycles of lithium-ion batteries in a controlled laboratory environment at a temperature of approximately 25°C (77°F), with each dataset including the following parameters: State of Charge (SoC) (%), Open-Circuit Voltage (OCV) (V), Current (A), Duration of discharge time (s) Battery Capacity (Ah) The following shows the data collection steps:

1. **Charge:** Charge the lithium-ion battery (manufacturer-rated nominal capacity = 3500 mAh) fully to 4.2 V at a controlled temperature of 25°C.
2. **Rest:** Allow the battery to rest for 1 hour to stabilize after charging.
3. **Discharge:** Discharge the battery at a constant current of 6.8 mA (C-rate of approximately 0.002C) for 5 minutes. This corresponds to about 10% of the battery's capacity.
4. **Rest:** Allow the battery to rest for 1 hour before recording its open-circuit voltage (OCV).
5. **Repeat:** Repeat steps 3 and 4 until the battery voltage reaches a cut-off level of 2.5 V. [6], [7]

B. Data Preprocessing

1. **Battery Capacity**: The effective battery capacity is computed by integrating the discharge current over time.

$$C_{\text{eff}} = \int_{t_0}^{t_f} I(t)\, dt \tag{1}$$

2. **Initial State of Charge**: Based on the initial voltage measurement, the initial SoC was determined via linear interpolation between 2.5V(0%) and 4.2V(100%).

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3. **Capacity Tracking**: For each discharge step, the change in capacity was computed from the current and discharge duration, then subtracted from or added to the current capacity depending on whether the battery was discharging or charging.

4. **SoC calculation**: The SoC was updated using the formula:

$$\text{SoC} = \frac{\text{Current Capacity}}{\text{Battery Capacity}} \times 100\%. \tag{2}$$

C. Kalman Filter Design

In this work, KF is used to estimate the battery's open circuit voltage (OCV) in real time. [8] Unlike a basic two-state model, our design extends the state vector to include the state of charge (SoC) for added adaptability. The filter is defined by the following elements:

1) **State Vector ($\mathbf{x}$):**

$$\mathbf{x} = \begin{bmatrix} \text{OCV} \\ \dot{\text{OCV}} \\ \text{SoC} \end{bmatrix} \tag{3}$$

The first component represents the battery's current OCV, the second represents the rate of change of OCV, and the third tracks the battery's state of charge. Including SoC allows the filter to dynamically adjust and bound the charge level during operation.

2) **State Transition Model ($\mathbf{F}$) and Control Input ($\mathbf{B}$):**

$$\mathbf{x}_{k+1} = \underbrace{\begin{bmatrix} 1 & 1 & k \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{\mathbf{F}} \mathbf{x}_k + \underbrace{\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}}_{\mathbf{B}} u_k + \mathbf{w}_k \tag{4}$$

- k is a coupling factor that quantifies the influence of SoC on the OCV dynamics.
- u_k represents the control input applied to the SoC (for example, a discharge rate).
- $\mathbf{w}_k$ is the process noise, assumed to be zero-mean Gaussian with covariance $\mathbf{Q}$.

3) **Measurement Model ($\mathbf{z}$):**

$$\mathbf{z}_k = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{\mathbf{H}} \mathbf{x}_k + \mathbf{v}_k \tag{5}$$

- The first row extracts the OCV, while the second row provides the SoC measurement.
- $\mathbf{v}_k$ is the measurement noise, assumed to be zero-mean Gaussian with covariance $\mathbf{R}$.

4) **Noise Covariances ($\mathbf{Q}$ and $\mathbf{R}$):**

- The process noise covariance is initialized as:

$$\mathbf{Q} = \begin{bmatrix} 1 \times 10^{-4} & 0 & 0 \\ 0 & 1 \times 10^{-3} & 0 \\ 0 & 0 & 1 \times 10^{-4} \end{bmatrix}, \tag{6}$$

and is dynamically updated using an exponentially weighted moving average (EWMA) of recent residual magnitudes with a smoothing factor (e.g., $\alpha_Q = 0.05$).

- The measurement noise covariance is set as:

$$\mathbf{R} = \begin{bmatrix} 1 \times 10^{-3} & 0 \\ 0 & 1 \times 10^{-2} \end{bmatrix}. \tag{7}$$

Although $\mathbf{R}$ can be updated based on the standard deviation of recent residuals, we keep it fixed (using $\alpha_R = 0$) for stability.

5) **Predict–Update Cycle:**

- **Predict Step:**

$$\mathbf{x}_{k|k-1} = \mathbf{F}\,\mathbf{x}_{k-1|k-1} + \mathbf{B}\,u_k, \tag{8}$$

$$\mathbf{P}_{k|k-1} = \mathbf{F}\,\mathbf{P}_{k-1|k-1}\,\mathbf{F}^\top + \mathbf{Q} \tag{9}$$

The state and its covariance $\mathbf{P}$ are projected forward based on the system dynamics.

- **Update Step:**

$$\mathbf{y}_k = \mathbf{z}_k - \mathbf{H}\,\mathbf{x}_{k|k-1}, \quad \mathbf{S}_k = \mathbf{H}\,\mathbf{P}_{k|k-1}\,\mathbf{H}^\top + \mathbf{R}, \tag{10}$$

$$\mathbf{K}_k = \mathbf{P}_{k|k-1}\,\mathbf{H}^\top\,\mathbf{S}_k^{-1}, \tag{11}$$

$$\mathbf{x}_{k|k} = \mathbf{x}_{k|k-1} + \mathbf{K}_k\,\mathbf{y}_k, \tag{12}$$

$$\mathbf{P}_{k|k} = (\mathbf{I} - \mathbf{K}_k\,\mathbf{H})\,\mathbf{P}_{k|k-1}\,(\mathbf{I} - \mathbf{K}_k\,\mathbf{H})^\top + \mathbf{K}_k\,\mathbf{R}\,\mathbf{K}_k^\top \tag{13}$$

The innovation $\mathbf{y}_k$ (the difference between the measured and predicted outputs) and the Kalman Gain $\mathbf{K}_k$ are used to correct the state estimate.

6) **Training and Testing:**

- Instead of stratified or random sampling, our approach partitions data on a per-battery basis: Out of eight cycles, the last two cycles (25%) of each battery are held out for testing, while the remaining cycles (75%) are used for training. This ensures that the temporal progression is respected.

7) **Residual Analysis & Adaptive Tuning:**

- After each update, the residual $\mathbf{y}_k$ is recorded. These residuals inform the adaptive tuning of $\mathbf{Q}$ (and potentially $\mathbf{R}$) via an EWMA. This adaptive mechanism helps the filter adjust to changes in noise levels and system behavior over time.

Initial State Note: In our implementation, the initial state is set using the first training measurements of OCV and SoC. Notably, the initial OCV rate is set to -10 to account for the battery discharge occurring during data collection.

Summary: The implemented Kalman Filter not only predicts the battery's OCV through its state evolution but also refines the estimates using new measurements of both OCV and SoC. By extending the state vector and incorporating adaptive noise tuning, the filter is well-equipped to handle the dynamic behavior observed in battery systems while respecting the natural temporal order of the data.

D. Polynomial Regression

A polynomial model was fitted to the OCV-SoC data to represent the relationship between open-circuit voltage (OCV) and state of charge (SoC). [9] The general equation for the polynomial is:

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$$\text{OCV}(\text{SoC}) = a_n \cdot \text{SoC}^n + a_{n-1} \cdot \text{SoC}^{n-1} + \cdots + a_1 \cdot \text{SoC} + a_0, \tag{14}$$

where $a_n, a_{n-1}, \ldots, a_0$ are the coefficients obtained through least squares fitting, and n is the degree of the polynomial.

a) Model Training: The complete dataset was partitioned on a per-battery basis by holding out the last two cycles for testing and using the remaining cycles for training. Polynomials of degrees 1 through 5 were fitted to the training data. The optimal degree was selected by minimizing the mean squared error (MSE) on the training set.

b) Prediction: The best polynomial model (determined to be degree 5) was applied to the test set to evaluate its performance. Metrics such as MSE and the determination coefficient (R^2) were calculated to compare the precision of the polynomial regression against the Kalman filter in subsequent sections.

c) Model Extraction: Using data from all batteries—with the last two cycles held out for testing—a global polynomial regression was performed on the training data (all but the last two cycles per battery). The fitted polynomial equation for V_{OC} as a function of SoC is given by:

$$\begin{aligned} V_{\text{OC}}(\text{SoC}) = {} & 6.634 \times 10^{-10} \cdot \text{SoC}^5 - 2.012 \times 10^{-7} \cdot \text{SoC}^4 \\ & + 2.221 \times 10^{-5} \cdot \text{SoC}^3 - 0.001107 \cdot \text{SoC}^2 \\ & + 0.03365 \cdot \text{SoC} + 3.163. \end{aligned} \tag{15}$$

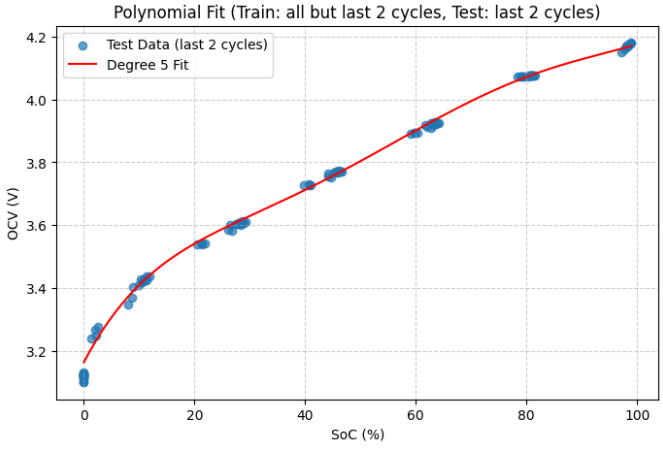


Fig. 1. Global Polynomial Fit of the OCV–SoC Relationship. The red curve represents the fitted model, while the scatter points correspond to the test data.

III. Results and Discussion

A. Overall Performance Comparison

The mean square error (MSE), root mean square error (RMSE), and coefficient of determination (R^2) were measured for both Kalman Filter (KF) and Polynomial Regression (PR) approaches. Both models were applied to predict the open circuit voltage (OCV) of the battery using the same dataset (with per-cell train-test splits). The consolidated train and test results are presented in Table I.

TABLE I
Consolidated Metrics Table

Model	Data	MSE	RMSE	R^2
Kalman Filter	Train	1.32×10^{-6}	1.15×10^{-3}	0.99998884
Kalman Filter	Test	3.10×10^{-7}	5.57×10^{-4}	0.99999752
Polynomial	Train	2.57×10^{-4}	1.60×10^{-2}	0.997823
Polynomial	Test	4.02×10^{-4}	2.01×10^{-2}	0.996786

Key Observations

The key findings from the model comparisons reveal that the Kalman Filter (KF) significantly outperformed the Polynomial Regression (PR) model, achieving an RMSE as low as 5.57×10^{-4} V on the test data, while the PR model, though still providing a reasonable fit, exhibited higher error levels with an RMSE of 2.01×10^{-2} V on the test data, indicating lower precision compared to KF. The KF demonstrated near-perfect accuracy with R^2 values of 0.99998884 (Train) and 0.99999752 (Test), whereas the PR model achieved R^2 values of 0.997823 (Train) and 0.996786 (Test).

For the training results, the KF model achieved a training MSE of 1.32×10^{-6} and an RMSE of 1.15×10^{-3} V, indicating extremely low errors and a very precise fit on the training data, while in contrast, the PR model exhibited a higher training MSE of 2.57×10^{-4} and an RMSE of 1.60×10^{-2} V, suggesting that its fit on the training data was less accurate. On the test data, the KF model maintained its high performance with an MSE of 3.10×10^{-7} and an RMSE of 5.57×10^{-4} V, demonstrating excellent generalization to unseen data, whereas the PR model showed a notable increase in error on the test data with an MSE of 4.02×10^{-4} and an RMSE of 2.01×10^{-2} V, highlighting its relatively lower robustness.

B. Per-Cell (Per-Battery) Performance Comparison

The results of the 8 cells are compared to analyze the performance of Kalman Filter and Polynomial. The detailed train and test results are presented in Table II and Table III.

a) Key Observations:

- **Kalman Filter (KF) Consistency & Accuracy:**
  - Across all eight batteries, the KF consistently yields very low MSE and RMSE on both training and testing data (on the order of $\sim 10^{-6}$ for training MSE and $\sim 10^{-7}$ for testing MSE).
  - The near-unity R^2 values (≈ 0.99999–0.999998) indicate that the KF captures the OCV–SoC relationship exceptionally well and generalizes effectively to unseen test cycles.
- **Polynomial Regression (PR) Performance:**
  - While the PR model also demonstrates reasonable accuracy, its MSE and RMSE values are notably higher than those of the KF for most batteries, especially in the test set.
  - The best PR results remain acceptable for many applications, but show greater variation (e.g., Bat-

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TABLE II
PER-CELL (PER-BATTERY) PERFORMANCE ON TRAINING DATA

Battery	Model	Train MSE	Train RMSE	Train R^2
1	KF	1.31×10^{-6}	1.14×10^{-3}	0.999990
1	Polynomial	4.46×10^{-4}	2.11×10^{-2}	0.996550
2	KF	1.30×10^{-6}	1.14×10^{-3}	0.999988
2	Polynomial	1.21×10^{-4}	1.10×10^{-2}	0.998927
3	KF	1.30×10^{-6}	1.14×10^{-3}	0.999989
3	Polynomial	1.24×10^{-4}	1.11×10^{-2}	0.998903
4	KF	1.30×10^{-6}	1.14×10^{-3}	0.999989
4	Polynomial	2.96×10^{-4}	1.72×10^{-2}	0.997576
5	KF	1.49×10^{-6}	1.22×10^{-3}	0.999988
5	Polynomial	5.70×10^{-4}	2.39×10^{-2}	0.995557
6	KF	1.30×10^{-6}	1.14×10^{-3}	0.999989
6	Polynomial	1.34×10^{-4}	1.16×10^{-2}	0.998813
7	KF	1.29×10^{-6}	1.13×10^{-3}	0.999989
7	Polynomial	2.60×10^{-4}	1.61×10^{-2}	0.997684
8	KF	1.30×10^{-6}	1.14×10^{-3}	0.999989
8	Polynomial	1.61×10^{-4}	1.27×10^{-2}	0.998597

TABLE III
PER-CELL (PER-BATTERY) PERFORMANCE ON TEST DATA

Battery	Model	Test MSE	Test RMSE	Test R^2
1	KF	3.00×10^{-7}	5.48×10^{-4}	0.999998
1	Polynomial	3.32×10^{-4}	3.32×10^{-4}	0.997541
2	KF	3.10×10^{-7}	5.57×10^{-4}	0.999997
2	Polynomial	2.91×10^{-4}	2.91×10^{-4}	0.997558
3	KF	3.09×10^{-7}	5.56×10^{-4}	0.999997
3	Polynomial	3.25×10^{-4}	3.25×10^{-4}	0.997302
4	KF	3.07×10^{-7}	5.54×10^{-4}	0.999998
4	Polynomial	7.08×10^{-4}	7.08×10^{-4}	0.994527
5	KF	3.24×10^{-7}	5.69×10^{-4}	0.999998
5	Polynomial	4.58×10^{-4}	4.58×10^{-4}	0.996616
6	KF	3.11×10^{-7}	5.57×10^{-4}	0.999997
6	Polynomial	3.27×10^{-4}	3.27×10^{-4}	0.997261
7	KF	3.07×10^{-7}	5.54×10^{-4}	0.999997
7	Polynomial	3.29×10^{-4}	3.29×10^{-4}	0.997214
8	KF	3.12×10^{-7}	5.58×10^{-4}	0.999997
8	Polynomial	4.46×10^{-4}	4.46×10^{-4}	0.996312

tery 4's test MSE is 7.08×10^{-4}, noticeably larger than for other cells).

- **Minimal Variation Across Batteries for KF:**
  - The KF model's metrics remain extremely consistent across all eight cells, suggesting that it is robust to cell-to-cell differences (such as slight variations in chemistry or aging).
  - By contrast, the PR model sees somewhat larger variability from cell to cell (again, Battery 4 stands out with the highest test MSE).
- **Implications for Real-Time Estimation:**
  - The KF's strong performance (low MSE/RMSE and high R^2) on both train and test data indicates it can provide reliable real-time OCV estimates.
  - The PR model, while simpler to implement, may require more careful calibration to match the KF's level of accuracy, especially if used across a range of different battery cells.

b) Example of the test results by cell: Table IV and Table V shows a snippet of the generated test results by Kalman Filter(KF) and Polynomial Regression (PR).

TABLE IV
SNIPPET OF BATTERY 1 DATA SHOWING ACTUAL VS. KF-PREDICTED OCV

Battery	OCV(V)	SoC(%)	KF Predicted OCV (V)	KF Error (V)
1	4.177	98.65	4.1766	0.00044
1	4.075	79.44	4.0763	-0.00127
1	3.893	60.23	3.8933	-0.00027
1	3.728	41.02	3.7279	0.00011
1	3.541	21.81	3.5408	0.00025

TABLE V
SNIPPET OF BATTERY 1 DATA SHOWING ACTUAL VS. PR-PREDICTED OCV

Battery	OCV(V)	SoC(%)	PR Predicted OCV (V)	PR Error (V)
1	4.177	98.65	4.1692	0.00781
1	4.075	79.44	4.0682	0.00680
1	3.893	60.23	3.9040	-0.01101
1	3.728	41.02	3.7210	0.00703
1	3.541	21.81	3.5586	-0.01761

C. Kalman Filter and Polynomial Regression Performance with Added Noise

Real-world battery measurements are inherently noisy, which can lead to inaccurate SoC estimates or suboptimal charging/discharging in BMS. [10] To emulate these conditions, Gaussian noise (standard deviation $= 0.01$) was introduced into both the training and test data for both models.

a) Noise Handling.:

- **Kalman Filter (KF).** Noise is added at each measurement update, reflecting an online scenario where each new reading is imperfect. Because the KF is run per-battery, it can adapt its process noise covariance ($\mathbf{Q}$) for each cell independently.
- **Polynomial Regression (PR).** A global polynomial model is fit once on a noisy training set aggregated from all batteries. While simpler, this approach lacks the KF's per-battery adaptiveness and may be more susceptible to variations or outliers in any one cell's data.

Global vs. Per-Battery Approach. Although both methods use the same last-two-cycle test split, the KF's per-battery adaptation can yield different behaviors from the global polynomial approach. Table VI compares their MSE, RMSE, and R^2 under noisy conditions.

TABLE VI
KF AND PR PERFORMANCE WITH NOISE IN TRAINING AND TEST DATA

Model	Data Split	MSE	RMSE	R^2
KF	Train	7.31×10^{-5}	0.00855	0.999382
KF	Test	8.55×10^{-5}	0.00924	0.999317
PR	Train	2.58×10^{-4}	0.01608	0.997815
PR	Test	4.16×10^{-4}	0.02040	0.996672

Key Observations: As expected, noise increases both MSE and RMSE relative to Table I. However, KF maintains $R^2 \approx$

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0.999 through adaptive $\mathbf{Q}$ tuning, while PR achieves $R^2 \approx 0.997$ but cannot adapt online to varying noise levels. These findings confirm KF's superiority for real-time estimation under noisy conditions, though PR remains viable for offline applications with stable measurements.

D. Residual Analysis for Training and Testing

Residuals provide important insights into the model's performance, such as spotting overfitting and evaluating the distribution of the data points. The residuals are calculated as:

$$\text{Residual} = z - H.x \tag{16}$$

Figs 2 and 3 show the training and test residuals for the Kalman Filter and Polynomial Regression model.



Fig. 2. Kalman Filter and Polynomial Residuals on Training Set. The blue points denote the KF training residuals, while the orange points denote the PR training residuals.

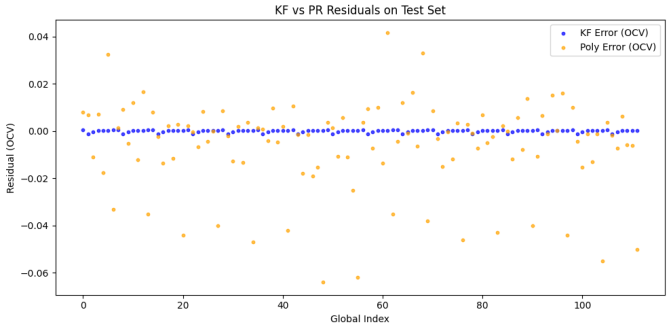


Fig. 3. Kalman Filter and Polynomial Residuals on Testing Set. The blue points denote the KF testing residuals, while the orange points denote the PR testing residuals.

Residual plots (Figs. 2–3) reveal key differences between models. KF residuals cluster tightly around zero in both training and test sets, indicating consistent accuracy. PR residuals show wider spread, particularly on test data where they drift noticeably from zero. This suggests PR may overfit the training data or lack sufficient flexibility to generalize, confirming KF's superior stability for OCV prediction.

IV. Conclusion

Accurate OCV estimation is vital for lithium-ion battery management systems. This comparative study demonstrates that the Kalman Filter achieves superior accuracy (RMSE $\sim 10^{-4}$V) through adaptive noise tuning, making it ideal for real-time applications. Polynomial Regression, while less accurate, offers simplicity for offline scenarios where computational efficiency outweighs adaptability.

Future work should extend these methods to varied temperatures, battery chemistries, and aging conditions to enhance robustness. [1], [11] The choice between KF and PR ultimately depends on application-specific requirements for accuracy, computational resources, and real-time performance.

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References

[1] C. Wang, R. Li, Y. Cao, and M. Li, "A hybrid model for state of charge estimation of lithium-ion batteries utilizing improved adaptive extended Kalman filter and long short-term memory neural network," *J. Power Sources*, vol. 620, p. 235272, 2024, doi: 10.1016/j.jpowsour.2024.235272.

[2] F. Elmahdi, L. Ismail, and M. Noureddine, "Fitting the OCV-SOC relationship of a battery lithium-ion using genetic algorithm method," *E3S Web Conf.*, vol. 234, p. 00097, Jan. 2021, doi: 10.1051/e3sconf/202123400097.

[3] L. Mao, Q. Hu, J. Zhao, and X. Yu, "State-of-charge of lithium-ion battery based on equivalent circuit model – Relevance vector machine fusion model considering varying ambient temperatures," *Measurement*, vol. 221, p. 113487, 2023, doi: 10.1016/j.measurement.2023.113487.

[4] D. Ma, K. Gao, Y. Mu, Z. Wei, and R. Du, "An adaptive tracking-extended Kalman filter for SOC estimation of batteries with model uncertainty and sensor error," *Energies*, vol. 15, no. 10, p. 3499, 2022, doi: 10.3390/en15103499.

[5] A. Tabine, E. M. Laadissi, A. Elachhab, S. Bouzaid, C. Ennawaoui, and A. Hajjaji, "A novel fitting polynomial approach for an accurate SOC estimation in Li-ion batteries considering temperature hysteresis," *e-Prime – Adv. Electr. Eng., Electron. Energy*, vol. 10, p. 100822, 2024, doi: 10.1016/j.prime.2024.100822.

[6] P. Pillai, S. Sundaresan, P. Kumar, K. R. Pattipati, and B. Balasingam, "Open-circuit voltage models for battery management systems: A review," *Energies*, vol. 15, no. 18, p. 6803, Sep. 2022, doi: 10.3390/en15186803.

[7] Center for Advanced Life Cycle Engineering, "Battery research data," UMD, 2023. [Online]. Available: https://calce.umd.edu/data [Accessed: 3 Mar. 2025].

[8] M. Hossain, M. E. Haque, and M. T. Arif, "Kalman filtering techniques for the online model parameters and state of charge estimation of Li-ion batteries: A comparative analysis," *J. Energy Storage*, vol. 51, p. 104174, 2022, doi: 10.1016/j.est.2022.104174.

[9] M. Chen and G. A. Rincón-Mora, "An accurate electrical battery model capable of predicting runtime and I–V performance," *IEEE Trans. Energy Convers.*, vol. 21, no. 2, pp. 504–511, Mar. 2005. [Online]. Available: https://rincon-mora.gatech.edu/publicat/jrnls/tec05$_b att_m dl.pdf$

[10] I. Marocco, "How measurement systems can improve EV/HEV batteries in noisy environments," *EDN Asia*, 2020. [Online]. Available: https://www.ednasia.com/how-measurement-systems-can-improve-ev-hev-batteries-in-noisy-environments/ [Accessed: 3 Mar. 2025].

[11] C. Zhao and X. Li, "Hierarchical deep learning model for degradation prediction per look-ahead scheduled battery usage profile," *IEEE Trans. Smart Grid*, accepted for publication, Sep. 2024.

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